ParkiScan Neuroverse

Game + Digital Twin for freezing of gait in Parkinson's disease

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Digital Twin Challenge · Phase 1 submission

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Research prototype · not a medical device · synthetic or anonymised open data only

Problem statement and objective

PROBLEM

- Parkinson's disease: about 5.8 lakh people in India, roughly 10% of the global burden.
- Freezing of gait, a sudden inability to step, affects about half of patients in earlier stages and up to 80% as the disease progresses. It is a leading cause of falls.
- Care is reactive. A neurologist sees the patient every few months and relies on recall, while freezing, wearing-off and falls happen at home in between.

OBJECTIVE

- Build a digital twin of one patient that fuses a synthetic health record, wearable streams, a levodopa model and a home game.
- Predict the adverse event, freezing of gait: seconds ahead as a warning, and hours ahead as a risk window.
- Give the clinician a dashboard to see risk and its causes, and to test what-if changes before the next visit.

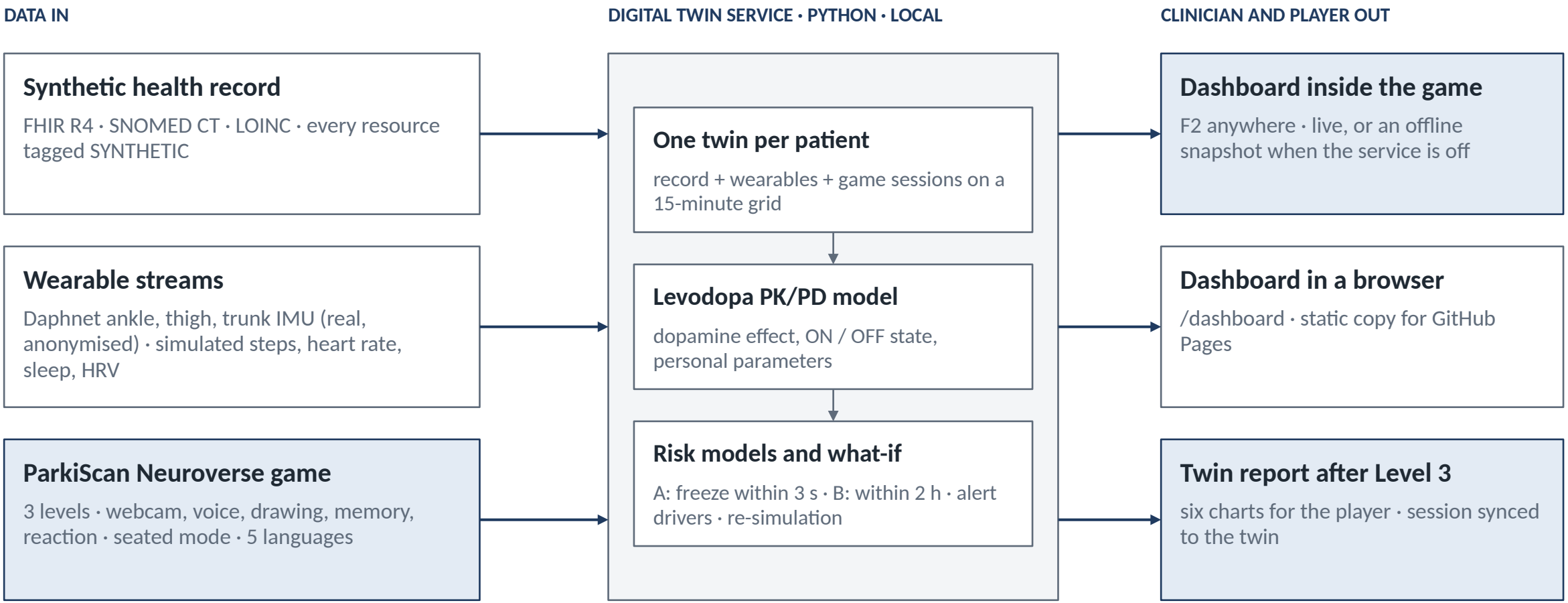
Use case. A person with Parkinson's plays a short game on a laptop and may wear a low-cost ankle sensor. After each session the twin updates, forecasts freezing risk for the next hours and lets the clinician try changes on the twin first.

Sources: Khurana & Gourie-Devi, Cureus 2025 (India prevalence); Conde et al., Front Neurol 2023 (freezing and falls).

Challenge requirements and how we meet them

Requirement	What we built	Where
Digital twin of a patient with a localised adverse event	One virtual Parkinson's patient; the adverse event is freezing of gait	Twin service
Static data: synthetic EHR	FHIR R4 record with SNOMED CT and LOINC codes, tagged SYNTHETIC; real clinical header for the 10 Daphnet participants	ParkiScanML/parkiscan/twin
Dynamic data: wearables / IoT	Real Daphnet ankle, thigh and trunk accelerometers; simulated steps, heart rate, sleep, HRV; ESP32 sensor path	Daphnet, simulator, firmware
Fuse both streams and predict the event	Model A: freeze within 3 s, AUC 0.76 on real data. Model B: freeze within 2 h, AUC 0.806 on a synthetic cohort	Reports in ParkiScanML/reports
Doctor dashboard	Risk timeline, alert drivers, ON / OFF curve, what-if, sensor replay, record, model card. Opens inside the game (F2) and in a browser	Game, /dashboard
Sandbox data rules	Only synthetic or anonymised open data; everything runs and stays on the computer	Whole project
Submission package	GitHub README, Apache-2.0 licence, demo video of 20 minutes or more, architecture PDF, presentation PDF	Repository, docs/

Architecture



Part of the ParkiScan Neuroverse game

Data source or output outside the game

All data stays on the computer

End-to-end flow

1

Play

The patient plays three levels of short tasks. Seated mode and a clean skip are always available.

2

Measure

Each task is recorded on the computer: timing, accuracy, smoothness and tracking quality.

3

Sync

At the end of Level 3 the session goes to the local twin service and becomes a patient.

4

Fuse

The twin joins record, wearables, the levodopa curve and game measures on a 15-minute grid.

5

Predict

Model A: freeze within 3 s. Model B: freeze within 2 h, with the drivers behind each alert.

6

Review

The clinician opens the dashboard in the game: risk, alerts, what-if, replay, record, model card.

If the twin service is off

The in-game dashboard opens a bundled snapshot of 16 twins with 75 what-if scenarios each. The latest session stays listed and can be synced later.

Closing the loop

The clinician tries a change on the twin, such as shifting a dose time, discusses it with the patient, and the next session updates the twin. What-if output is for discussion, not a dosing instruction.

What the game extracts

Level · task	Measures extracted	Domain
1 · Follow the memory lantern (webcam gaze)	pointer error, % on target, re-acquisition latency, flash reaction time, eye-calibration error	Visual attention
1 · Memory Core: sustained “Ahh”	voiced duration, pitch mean and variability, jitter and shimmer proxies, HNR	Voice
1 · Memory Core: “pa-ta-ka” and sentence	onset rate and regularity, pauses, speech rate	Speech motor
1 · Memory Core: clap to 90 BPM	inter-clap interval and variability, tempo error, asynchrony to the beat	Rhythm timing
1 · Recover the lost memory	hits, omissions, false recognitions, recall time	Memory
2 · Trace the circuit (spiral)	radial deviation, tremor-band proxy, smoothness, pen lifts	Fine motor, tremor
2 · Connect nodes, repair component	completion time, wrong connections, placement and rotation error	Sequencing, visuospatial
2 · Reactor signal	reaction time mean and SD, false and missed responses	Reaction time
3 · Mirror the avatar (webcam pose)	moves completed, arm elevation, similarity, tracking confidence	Upper-limb movement
3 · Glowing path (steps on the spot)	steps, cadence, step-interval SD, stop and go latency	Stepping, stopping
3 · Memory vehicle (seated steering)	lane offset, time in lane, steering steadiness, brake and go latency	Hand reactions
3 · Six check-ins	answers per domain; kept separate and never scored	Self-report

How it is used. Every record carries quality flags (valid, caveats, insufficient, invalid); QA and simulated runs are labelled. On the twin the measures sit beside the levodopa ON / OFF curve. They are not yet inputs to Models A and B, because no clinician-labelled game sessions exist to train on.

Predictions and clinician dashboard

MODEL A · REAL DAPHNET DATA · FREEZE WITHIN 3 S

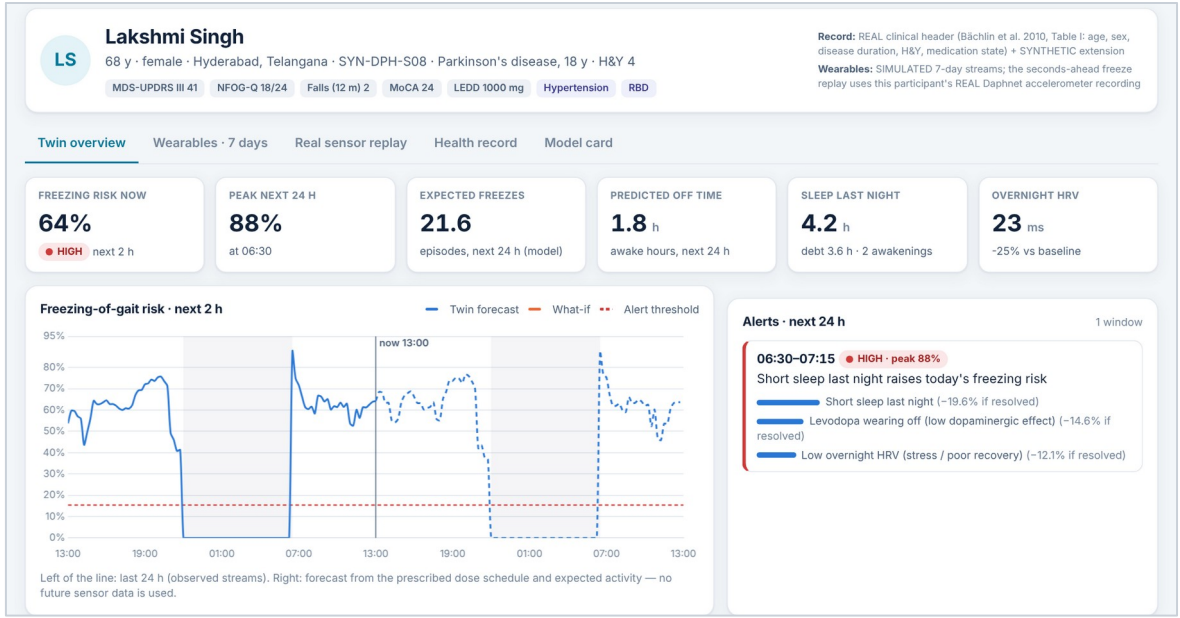
Freeze-index baseline, AUC	0.60
Ankle IMU model, AUC (95% CI)	**0.76 (0.72–0.83)**
Freezes warned before onset	62% (148 of 237)
Median warning lead time	4.6 s
False cues per hour of walking	96
Conservative setting	43% warned at 40 false cues / h

Leave-one-subject-out on 10 people; the test person is never seen in training.

MODEL B · SYNTHETIC COHORT · FREEZE WITHIN 2 H

Health record only, AUC	0.779
Wearables only, AUC	0.759
Record + wearables, AUC	0.799
Digital twin (+ levodopa model), AUC	**0.806**

Top 10% of alerts capture 45% of freeze windows (lift 4.5×). Labels come from a documented simulator: this shows the fusion works end to end, not clinical accuracy.



Clinician dashboard: freezing risk now and for the next 24 h, alerts with their causes, levodopa ON / OFF curve and what-if simulator. The same views open inside the game with F2.

Novelty, limits and next steps

WHAT IS NEW

A game that is also the twin's data source. The patient plays; the clinician view opens inside the same program.

Four sources in one twin. Record, wearables, game and a personal levodopa PK/PD model.

Seconds-ahead forecast, honestly scored. Leave-one-subject-out, with warned %, lead time and false cues per hour instead of one AUC.

Alerts explain themselves. Each driver is removed in simulation to show how much risk it adds; what-if re-runs the twin with identical noise.

Built for Parkinson's. Webcam only, no hardware needed, seated mode, clean skip, no pass or fail, five languages, runs offline.

WHAT WE DO NOT CLAIM

- No diagnosis, staging or dosing advice.
- Model A: 10 people in a lab protocol that provokes freezing; home false-cue rates will differ.
- Model B and consumer-wearable streams are synthetic.
- Webcam measures are 2-D estimates, not clinical tracking.

NEXT STEPS

- Clinician-labelled game sessions, with ethics approval.
- Feed game measures into the models.
- Wearable cue pilot with the ESP32 sensor.
- Native-speaker review of translations.

Freezing of gait: from surprise to forecast, from one home game and one digital twin.